

① AND

$$a = 5 \Rightarrow 0101$$

$$b = 6 \Rightarrow 0110$$

a	b	a & b
0	0	0
1	1	1
0	1	0
1	0	0

32 16 8 4 2 1

0 1 0 0

$\Rightarrow 4$

②

$$a = 10 \Rightarrow 1010$$

$$b = 12 \Rightarrow 1100$$

a	b	a & b
1	1	1
0	1	0
1	0	0
0	0	0

32 16 8 4 2 1

1 0 0 0

$\Rightarrow 8$

③ $a = 15 \Rightarrow 1111$
 $b = 16 \Rightarrow 10000$

a	b	a & b
0	1	0
1	0	0
1	0	0
1	0	0
1	0	0

a	b	a & b
0	1	0
1	0	0
1	0	0
1	0	0
1	0	0

$\Rightarrow 0$

OR

① $a = 5 \Rightarrow 0101$
 $b = 6 \Rightarrow 0110$

a	b	a b
0	0	0
1	1	1
0	1	1
1	0	1

② $a = 10 \Rightarrow 1010$
 $b = 12 \Rightarrow 1100$

a	b	a b
1	1	1
0	1	1
1	0	1
0	0	0

③ $a = 13 \Rightarrow 1101$
 $b = 14 \Rightarrow 1110$

a	b	a b
1	1	1
1	1	1
0	1	1
1	0	1

32 16 8 4 2 1
 0 1 1 1
 $\Rightarrow 7$

$\Rightarrow 7$

XOR

① $a = 5 \Rightarrow 0101$
 $b = 6 \Rightarrow 0110$

a	b	a ^ b
0	0	0
1	1	0
0	1	1
1	0	1

② $a = 18 \Rightarrow 10010$
 $b = 19 \Rightarrow 10011$

a	b	a ^ b
1	1	0
0	0	0
0	0	0
1	1	0
0	0	1

③ $a = 22 \Rightarrow 10110$
 $b = 24 \Rightarrow 11000$

a	b	a ^ b
1	1	0
0	1	1
1	0	1
1	0	1
0	0	0

32 16 8 4 2 1
 0 0 1 1
 $\Rightarrow 3$

32 16 8 4 2 1
 0 0 0 1 1
 $\Rightarrow 6$

32 16 8 4 2 1
 0 1 1 1 0
 $\Rightarrow 14$

Bitwise left shift operator

1. $11 \ll 2$

8421
1011

2. $12 \ll 1$ $3 \ll 8 \ll 2$

16 8 4 2 1
0 1 1 0 0
1000
1000

32 16 8 4 2 1
1 1 0 0

$\Rightarrow 24$

$\Rightarrow 44$

Right shift operator

1. $16 \gg 3$

32 16 8 4 2 1
1 0 0 0 0
10

$0/p = 2$

2. $18 \gg 1$ 3. $10 \gg 3$

32 16 8 4 2 1
1 0 0 1 0
1001

32 16 8 4 2 1
1 0 1 1
1

$\Rightarrow 9$